Krishna Durgavarjhula

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY August 2026 - May 2030 (expected)
Bachelor of Arts in Computer Science

AMERICAN HIGH SCHOOL August 2022 - May 2026
High School Diploma, GPA: 3.74 (unweighted)
Relevant Coursework: AP Computer Science A (AP CSA), AP Physics 1, AP Calculus BC, AP Physics C (Mechanics and E&M), AP Government, AP Economics
Clubs: Computer Science Club (Secretary & Instructor), Quiz Bowl (Division C Team Member)
Test Scores: AP CSA: 5; AP Physics 1: 4; AP U.S. History: 4; AP World History: 4; AP CSP: 4; SAT: 1510

Other Relevant Coursework: Intro to C++ Programming, 2D Design, AP Computer Science Principles (APCSP)

SKILLS

- **Concepts:** Object-Oriented Programming, Design Principles, Autonomous Systems, Project Management, Debugging and Testing, Data Structures and Algorithms
- **Languages**: Java, Python, C++, HTML/CSS/JavaScript, MATLAB
- **Databases/Frameworks/Libraries/Tools**: Node.js, NumPy, TensorFlow, Keras, React, Claude Code, Codex, Antigravity

PUBLICATIONS

Arora, J.S., Durgavarjhula, K., & Bansal, R. (2025). An Assistive Sensory-Based System for Integrated Fall and Obstacle Detection. 2025 3rd International Conference on Computer System, Information Technology, and Electrical Engineering (COSITE), 37-42.

WORK EXPERIENCE

LIFT AND SHIFT FOUNDATION | *Web Design Intern* January 2023 - Present
- Designed and developed interactive UI features on the Lift and Shift website that teaches block coding and 3D modeling for veterans re-entering the workforce.
- Gained experience with HTML/CSS/JS, GitHub, collaborative development, and web architectures.

PROJECT EXPERIENCE

UC BERKELEY ROAR ACADEMY | *Student* July 2025 - August 2025
- Performed machine learning simulations in Python for autonomous driving using deep neural networks, gradient descent, and reinforcement learning.
- Gained experience with Python libraries, AI/ML, and software simulation of real-world environments, working with UC Berkeley faculty and student researchers.

BREATHE APP | *Developer* July 2024 - August 2024
- Developed an Android app where users can log their respiratory symptoms and view daily weather data to understand their triggers over time.
- Utilized Javascript for native full-stack development, Node to deploy the backend, Firebase for dynamic user authentication, and Expo for iterative testing, and published the app to the Google Play store.

ALAMEDA COUNTY SCIENCE & ENGINEERING FAIR | *Team Member* August 2024 - December 2025
- Worked with a team of three to build an environmentally-friendly smart cane to assist seniors in avoiding obstacles and detecting falls along with a caretaker mobile app.
- Developed a system with ultrasonic sensors, a gyroscope, a Bluetooth Module, and an ergonomic handle design with a pulse sensor, haptic motors, and a thermoelectric generator that uses body heat to partially power the system.
- Developed C++ algorithms for ultrasonic terrain profiling, fall detection, and pulse tracking. Integrated via Bluetooth with the Android app, which displays user location, BPM, and fall alerts.
- Won 1st place category award in ACSEF, qualified for CSEF, and won 1st place team engineering project at STEM4ALL. Currently patent-pending, and presented and published at the IEEE-sponsored COSITE 2025.

BLUESTAMP ENGINEERING | *Student* July 2023 - August 2023
- Designed and built a phone-controlled robotic arm using servos, C++, and Arduino, controlled via a Bluetooth module using an app.
- Gained experience in digital electronics, hardware-software integration, project management, and iterative development.
- Prepared a presentation and demonstrated the project to an audience. Documented my work at kri-dur.github.io.

CODING TOMORROW INITIATIVE | *Instructor* June 2025 - July 2025
- Volunteered as an instructor teaching a class to 20 middle schoolers online at a non-profit.
- Implemented a web development curriculum with lesson plans, coding examples, and homework assignments.

AWARDS

Alameda County Science & Engineering Fair (ACSEF) 1st Place Category Award, STEM4ALL 1st Place Engineering Project (Team), California Science & Engineering Fair Certificate, BlueStamp Engineering Certificate Of Excellence, AP Scholar With Distinction, College Board School Recognition Award, National Merit Scholar Letter of Commendation